

Ist Iteration

Part 3 Jonathan Mugisha

1st Iteration

$$m = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Let $\alpha = 0.01$

$$\hat{y} = mx + b$$

$$= X \cdot m + b$$

$$= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \times \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1+6 \\ -4+20 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

Error = $\hat{y} - y = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

$$MSE = \frac{\sum (\hat{y} - y)^2}{n}$$

$$= \frac{\sum \begin{bmatrix} 1 \\ 11 \end{bmatrix}^2}{2} = \frac{\sum \begin{bmatrix} 1 \\ 121 \end{bmatrix}}{2} = \frac{1+121}{2} = \frac{122}{2} = 61$$

$$m_{new} = m_{old} - \alpha \frac{\partial MSE}{\partial m}$$

$$J = \frac{1}{n} \sum (y - \hat{y})^2 \quad \left| \frac{\partial J}{\partial \hat{y}} = \frac{2}{n} (\hat{y} - y) \right.$$

$$\frac{\partial J}{\partial \hat{y}} = \frac{2}{n} e = \frac{2}{2} e = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$\hat{y} = X m + b$$

$$\frac{\partial \hat{y}}{\partial m} = X$$

$$\frac{\partial J}{\partial m} = \frac{2}{n} X^T e$$

$$X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 1+44 \\ 3+110 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\frac{\partial J}{\partial m} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$m_{new} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - \begin{bmatrix} 0.45 \\ 1.13 \end{bmatrix}$$

$$m_{new} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$\frac{\partial J}{\partial b} = \frac{2}{n} e = \frac{2}{2} e = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.01 \\ 0.01 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 0.99 \end{bmatrix}$$

$$m_{\text{new}} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 0.99 \\ 0.99 \end{bmatrix}$$

x_1	x_2	y	$\hat{y}$
1	3	5	6
4	10	6	17

2nd Iteration

Notas

Harmony

18 Jueves /Thursday Beata Beatriz

Iteration 2

$$X_{m_1} = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} =$$

$$\begin{bmatrix} 1(-1.45) + 3(0.87) \\ 4(-1.45) + 10(0.87) \end{bmatrix}$$

$$= \begin{bmatrix} 1.16 \\ 2.9 \end{bmatrix}$$

19 Viernes /Friday San Canuto

$$\hat{y} = X_{m_1} + b_1$$

$$= \begin{bmatrix} 1.16 \\ 2.9 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$$

20 Sábado /Saturday San Fabián

21 Domingo /Sunday Santa Inés

$$e = \hat{y} - y$$

$$= \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

Enero /January

2024 Semana 04

22 Lunes /Monday San Vicente

$$J_{MSE} = \frac{1}{2} [(-2.85)^2 + (-2.21)^2]$$

$$= \frac{1}{2} (8.1225 + 4.8841)$$

$$= 6.5033$$

23 Martes /Tuesday San Idefonso

$$* \frac{dMSE}{dm} = \frac{2}{n} e$$

$$* \frac{dw}{db} = 1$$

$$* \frac{dw}{dm} = X^T$$

24 Miércoles /Wednesday San Francisco de Asís

$$\rightarrow \frac{dMSE}{dm} = \frac{2}{n} X^T e$$

$$\rightarrow \frac{dMSE}{db} = \frac{2}{n} e \times 1$$

$$\rightarrow X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix} = \begin{bmatrix} 1(-2.85) + 4(-2.21) \\ 3(-2.85) + 10(-2.21) \end{bmatrix}$$

$$= \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}$$

25 Jueves /Thursday

La conversión de San Pablo

$$m_{\text{new}} = m_{\text{old}} - \frac{\partial \text{MSE}}{\partial m}$$

$$= \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}$$

$$= \begin{bmatrix} 1.3331 \\ 1.1765 \end{bmatrix}$$

26 Viernes /Friday

Santos Timoteo y Tito
Adelanto de Sto. Tomás de Aquino

$$b_{\text{new}} = b_{\text{old}} - \frac{\partial \text{MSE}}{\partial b}$$

$$= \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} - 0.01 \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$$

$$= \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$$

27 Sábado /Saturday
Santa Ángela de Merici

28 Domingo /Sunday
Santo Tomás de Aquino, patrón de Psicólogos

3rd Iteration

King Obafemi Abeginin
3rd Iteration

$$m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}, x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

$$b = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}, y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}, \alpha = 0.01$$

$$\hat{y} = mx + b$$

$$\hat{y} = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 3.2149 \\ 7.3455 \end{bmatrix}$$

$$e = \hat{y} - y$$

$$= \begin{bmatrix} 3.2149 \\ 7.3455 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$$

$$MSE = \sum_{n=2}^n (\hat{y} - y)^2$$

$$= \frac{(-1.7851)^2 + (1.3447)^2}{2}$$

$$= \frac{3.1866 + 1.8082}{2}$$

$$= \frac{4.9948}{2} = 2.4974$$

$$\frac{dMSE}{dm} = \frac{2}{n} X^T e, n=2$$

$$= \frac{2}{2} X^T e = X^T e$$

$$= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$$

$$\frac{dMSE}{dm} = \begin{bmatrix} 3.5937 \\ 8.0917 \end{bmatrix}$$

$$m_{new} = m_{old} - \alpha \frac{dMSE}{dm}$$

$$= \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} - 0.01 \begin{bmatrix} 3.5937 \\ 8.0917 \end{bmatrix}$$

$$= \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} - \begin{bmatrix} 0.035937 \\ 0.080917 \end{bmatrix}$$

$$m_{new} = \begin{bmatrix} -1.3690 \\ 1.0956 \end{bmatrix}$$

$$\frac{dMSE}{db} = \frac{2}{n} e, n=2$$

$$\frac{dMSE}{db} = \frac{2}{2} e = e$$

$$= \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$$

Abeginin
King

$$b_{new} = b_{old} - \alpha \frac{dMSE}{db}$$

$$= \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} - 0.01 \begin{bmatrix} -1.7851 \\ 1.3447 \end{bmatrix}$$

$$= \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} - \begin{bmatrix} -0.017851 \\ 0.013447 \end{bmatrix}$$

$$b_{new} = \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix}$$

4th Iteration

Erster Iteration

4th Iteration

$$m = \begin{bmatrix} -1.3690 \\ 1.0956 \end{bmatrix} \quad b = \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix} \quad \alpha = 0.01$$

$$\hat{y} = xm + b$$

$$mx = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3690 \\ 1.0956 \end{bmatrix} = \begin{bmatrix} -1.3690 + 3.2868 \\ -5.4760 + 10.9560 \end{bmatrix} \\ = \begin{bmatrix} 1.9178 \\ 5.4800 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 1.9178 \\ 5.4800 \end{bmatrix} + \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix} \\ = \begin{bmatrix} 2.9542 \\ 6.3787 \end{bmatrix}$$

$$\text{Error} \Rightarrow e = \hat{y} - y = \begin{bmatrix} 2.9542 \\ 6.3787 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$e = \begin{bmatrix} -2.0458 \\ 0.3787 \end{bmatrix}$$

$$\text{MSE} \Rightarrow \frac{(-2.0458)^2 + (0.3787)^2}{2} \\ = \frac{4.1853 + 0.1434}{2}$$

$$\text{MSE} = 2.1644$$

$$\frac{d\text{MSE}}{dm} = \frac{2}{n} x^T e, \quad n = 2$$

$$= x^T e$$

$$x^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$\frac{d\text{MSE}}{dm} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.0458 \\ 0.3787 \end{bmatrix} = \begin{bmatrix} -2.0458 + 1.5148 \\ -6.1374 + 3.7870 \end{bmatrix}$$

$$\frac{d\text{MSE}}{dm} = \begin{bmatrix} -0.5310 \\ -2.3504 \end{bmatrix}$$

Other JRMK085

$$m_{\text{new}} = m_{\text{old}} - \alpha \frac{dMSE}{dm} = \begin{bmatrix} -1.3690 \\ 1.0956 \end{bmatrix} - 0.01 \begin{bmatrix} -0.5310 \\ -2.3504 \end{bmatrix}$$

$$= \begin{bmatrix} -1.3690 + 0.00531 \\ 1.0956 + 0.023504 \end{bmatrix}$$

$$m_{\text{new}} = \begin{bmatrix} -1.3637 \\ 1.1191 \end{bmatrix}$$

$$\frac{dMSE}{db} = \frac{2}{n} e = \frac{2}{2} \begin{bmatrix} -2.0458 \\ 0.3187 \end{bmatrix}$$

$$\frac{dMSE}{db} = \begin{bmatrix} -2.0458 \\ 0.3187 \end{bmatrix}$$

$$b_{\text{new}} = b_{\text{old}} - \alpha \frac{dMSE}{db} = \begin{bmatrix} 1.0364 \\ 0.8987 \end{bmatrix} - 0.01 \begin{bmatrix} -2.0458 \\ 0.3187 \end{bmatrix}$$

$$= \begin{bmatrix} 1.0364 + 0.02046 \\ 0.8987 - 0.00319 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 1.0569 \\ 0.8949 \end{bmatrix}$$

Trend Observation

The values for m and b keep changing throughout the iterations because the gradient descent algorithm keeps on updating them so as to reduce the error associated with the predictions. As a result of such changes, there is a reduction in the errors involved in the predictions since their values keep on converging towards those that give accurate predictions. This leads to a constant reduction of error from 61.0 during the 1st iteration to 2.16 during the 4th iteration